Contents

Description Generator Reference Document

Date: December 26, 2024 **Purpose**: Comprehensive reference for the description generator evaluation system

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Executive Summary

This document provides a complete reference of: - **201 synthetic card game rules** organized by complexity level - **Feature vocabulary**: 29 feature types the system can detect - **Text-to-feature mappings**: 30 keywords that map to semantic features - **Category-to-feature mappings**: 43 rule categories with acceptable feature types

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Part 1: Feature Vocabulary

The description generator detects these semantic features from card hands:

Card Composition Features

Feature	Description	Example Detection
`is_flush`	All cards share the same suit	"all hearts"
`is_uniform_color`	All cards share the same color	"all red cards"
`unique_suits`	Count of distinct suits (1-4)	"3 different suits"
`unique_colors`	Count of distinct colors (1-2)	"both colors present"
`unique_ranks`	Count of distinct ranks (1-6)	"4 different ranks"
`has_pair`	At least two cards share a rank	"pair of 7s"

Feature	Description	Example Detection
has_triple	At least three cards share a rank	"three queens"

Positional Features

Feature	Description	Example Detection
first_card	Properties of position 0	"starts with spade"
last_card	Properties of final position	"ends with red"
ends_same_suit	First and last share suit	"bookends same suit"
ends_same_color	First and last share color	"red at both ends"
first_half	Properties of positions 0-2	"left side. . . "
second_half	Properties of positions 3-5	"right side. . . "
halves_same_color	Both halves have same color set	"halves match color"
halves_same_suits	Both halves have same suit set	"halves match suits"

Ordering Features

Feature	Description	Example Detection
is_sorted	Ranks in non-decreasing order	"increasing ranks"
is_palindrome_suits	Suit sequence reads same both ways	"suit palindrome"
is_palindrome_colors	Color sequence reads same both ways	"color palindrome"

Aggregate Features

Feature	Description	Example Detection
sum_ranks	Total of all rank values	"sum is 24"
rank_spread	Max rank - min rank	"spread of 8"
max_rank	Highest rank value	"contains king"
min_rank	Lowest rank value	"contains ace"
suit_count	Count of a specific suit	"3 hearts"
color_count	Count of a specific color	"4 red cards"
rank_count	Count of a specific rank	"2 sevens"

Meta Features

Feature	Description
distinguishing	General distinguishing characteristic

Part 2: Text-to-Feature Mappings

When descriptions contain these keywords, they map to the corresponding features:

Text Keyword	Maps To Features
"same suit"	is_flush, unique_suits
"same color"	is_uniform_color, unique_colors
"pair"	has_pair
"triple"	has_triple
"three of a kind"	has_triple
"sorted"	is_sorted
"increasing order"	is_sorted
"decreasing"	is_sorted
"first card"	first_card
"last card"	last_card
"starts with"	first_card
"ends with"	last_card
"first and last"	ends_same_color, ends_same_suit
"sum"	sum_ranks
"rank sum"	sum_ranks
"total"	sum_ranks
"spread"	rank_spread
"unique"	unique_colors, unique_ranks, unique_suits
"different"	unique_colors, unique_ranks, unique_suits
"palindrome"	is_palindrome_colors, is_palindrome_suits
"symmetric"	is_palindrome_colors, is_palindrome_suits
"halves"	halves_same_color
"hearts"	first_card, last_card, suit_count
"diamonds"	first_card, last_card, suit_count
"clubs"	first_card, last_card, suit_count
"spades"	first_card, last_card, suit_count
"red"	color_count, is_uniform_color
"black"	color_count, is_uniform_color
"face card"	max_rank, min_rank
"ace"	first_card, last_card, max_rank

Part 3: Category-to-Feature Mappings (Semantic Correctness)

For evaluation, each rule category maps to **acceptable features**. A description is "semantically correct" if it mentions ANY feature from the acceptable set.

Simple Categories (Specific Feature Sets)

Category	Acceptable Features
uniform	is_flush, is_uniform_color, unique_suits, unique_colors, unique_ranks
position_first	first_card
position_last	last_card
position_second	first_card
position_middle	first_card, last_card
terminals	first_card, last_card, ends_same_color, ends_same_suit
terminals_rank	first_card, last_card, rank_spread
sum	sum_ranks
sum_divisibility	sum_ranks

Category	Acceptable Features
`has`	`has_pair`, `has_triple`, `unique_ranks`
`no`	`has_pair`, `has_triple`, `unique_ranks`
`duplicates`	`has_pair`, `has_triple`, `unique_ranks`
`sorted`	`is_sorted`
`monotonic`	`is_sorted`, `rank_spread`
`alternating`	`is_uniform_color`, `unique_colors`
`palindrome`	`is_palindrome_colors`, `is_palindrome_suits`
`unique_count`	`unique_suits`, `unique_colors`, `unique_ranks`
`unique_bound`	`unique_suits`, `unique_colors`, `unique_ranks`
`count_specific`	`suit_count`, `color_count`, `rank_count`
`count_compare`	`suit_count`, `color_count`
`count_parity`	`suit_count`, `color_count`
`majority`	`suit_count`, `color_count`, `is_uniform_color`
`range`	`max_rank`, `min_rank`, `rank_spread`
`spread`	`max_rank`, `min_rank`, `rank_spread`
`adjacent`	`is_sorted`, `rank_spread`
`adjacent_constraint`	`is_sorted`, `unique_suits`, `unique_colors`
`runs`	`is_sorted`, `rank_spread`
`skip`	`is_sorted`, `rank_spread`
`position_pair`	`first_card`, `last_card`, `ends_same_suit`, `ends_same_color`
`shape`	`has_pair`, `has_triple`, `unique_ranks`
`halves_property`	`is_uniform_color`, `halves_same_color`
`halves_set`	`unique_suits`, `halves_same_color`
`halves_sum`	`sum_ranks`
`halves_copy`	`is_palindrome_suits`, `halves_same_color`
`periodic`	`unique_suits`, `unique_colors`

Lenient Categories (Accept ANY Feature)

These compositional categories are lenient because they may have multiple valid descriptions:

- `and_combination`, `or_combination`, `conditional`, `complex`, `xor`, `triple`, `negation`, `biconditional`

All accept the full feature set.

Part 4: Complete Rule Catalogue (201 Rules)

Level 1: Atomic Rules (50 rules)

Simple rules checking a single property.

Uniform Property Rules

ID	Name	Description	Category	Acceptable Features
atomic_001	All Same Suit	All cards have the same suit	uniform	`is_flush`, `unique_suits`, `unique_colors`, `unique_ranks`, `is_uniform_color`
atomic_002	All Same Color	All cards have the same color (red or black)	uniform	`is_flush`, `unique_suits`, `unique_colors`, `unique_ranks`, `is_uniform_color`

ID	Name	Description	Category	Acceptable Features
atomic_003	All Same Parity	All cards have the same rank parity (all odd or all even)	uniform	`is_flush`, `unique_suits`, `unique_colors`, `unique_ranks`, `is_uniform_color`
atomic_004	All Same Pointy/Round	All cards are pointy (spades/diamonds) or all are round (hearts/clubs)	uniform	`is_flush`, `unique_suits`, `unique_colors`, `unique_ranks`, `is_uniform_color`
atomic_005	All Same SH/DC	All cards are SH (spades/hearts) or all are DC (diamonds/clubs)	uniform	`is_flush`, `unique_suits`, `unique_colors`, `unique_ranks`, `is_uniform_color`

First Position Rules

ID	Name	Description	Category	Acceptable Features
atomic_006	First Card is Clubs	First card is a clubs	position_first	`first_card`
atomic_007	First Card is Diamonds	First card is a diamonds	position_first	`first_card`
atomic_008	First Card is Hearts	First card is a hearts	position_first	`first_card`
atomic_009	First Card is Spades	First card is a spades	position_first	`first_card`
atomic_010	First Card is Red	First card is red	position_first	`first_card`
atomic_011	First Card is Black	First card is black	position_first	`first_card`
atomic_012	First Card Rank is Odd	First card has odd rank	position_first	`first_card`
atomic_013	First Card Rank is Even	First card has even rank	position_first	`first_card`

Last Position Rules

ID	Name	Description	Category	Acceptable Features
atomic_014	Last Card is Clubs	Last card is a clubs	position_last	`last_card`
atomic_015	Last Card is Diamonds	Last card is a diamonds	position_last	`last_card`
atomic_016	Last Card is Hearts	Last card is a hearts	position_last	`last_card`
atomic_017	Last Card is Spades	Last card is a spades	position_last	`last_card`
atomic_018	Last Card is Red	Last card is red	position_last	`last_card`

ID	Name	Description	Category	Acceptable Features
atomic_019	Last Card is Black	Last card is black	position_last	`last_card`

Sum Rules

ID	Name	Description	Category	Acceptable Features
atomic_020	Sum of Ranks is Even	Sum of all rank values is even	sum	`sum_ranks`
atomic_021	Sum of Ranks is Odd	Sum of all rank values is odd	sum	`sum_ranks`
atomic_044	Sum of Ranks Divisible by 3	Sum divisible by 3	sum_divisibility	`sum_ranks`
atomic_045	Sum of Ranks Divisible by 4	Sum divisible by 4	sum_divisibility	`sum_ranks`
atomic_046	Sum of Ranks Divisible by 5	Sum divisible by 5	sum_divisibility	`sum_ranks`

Has/No Suit Rules

ID	Name	Description	Category	Acceptable Features
atomic_022	Has at Least One Clubs	Hand contains at least one clubs	has	`has_pair`, `has_triple`, `unique_ranks`
atomic_023	Has at Least One Diamonds	Hand contains at least one diamonds	has	`has_pair`, `has_triple`, `unique_ranks`
atomic_024	Has at Least One Hearts	Hand contains at least one hearts	has	`has_pair`, `has_triple`, `unique_ranks`
atomic_025	Has at Least One Spades	Hand contains at least one spades	has	`has_pair`, `has_triple`, `unique_ranks`
atomic_026	No Clubs	Hand contains no clubs	no	`has_pair`, `has_triple`, `unique_ranks`
atomic_027	No Diamonds	Hand contains no diamonds	no	`has_pair`, `has_triple`, `unique_ranks`
atomic_028	No Hearts	Hand contains no hearts	no	`has_pair`, `has_triple`, `unique_ranks`
atomic_029	No Spades	Hand contains no spades	no	`has_pair`, `has_triple`, `unique_ranks`

ID	Name	Description	Category	Acceptable Features
atomic_030	No Red Cards	Hand contains no red cards	no	`has_pair`, `has_triple`, `unique_ranks`
atomic_031	No Black Cards	Hand contains no black cards	no	`has_pair`, `has_triple`, `unique_ranks`

Middle Position Rules

ID	Name	Description	Category	Acceptable Features
atomic_032	Middle Card is Red	The middle card is red	position_middle	`first_card`, `last_card`
atomic_033	Middle Card is Black	The middle card is black	position_middle	`first_card`, `last_card`
atomic_034	Middle Card is Clubs	The middle card is a clubs	position_middle	`first_card`, `last_card`
atomic_035	Middle Card is Diamonds	The middle card is a diamonds	position_middle	`first_card`, `last_card`
atomic_036	Middle Card is Hearts	The middle card is a hearts	position_middle	`first_card`, `last_card`
atomic_037	Middle Card is Spades	The middle card is a spades	position_middle	`first_card`, `last_card`

Second Position Rules

ID	Name	Description	Category	Acceptable Features
atomic_038	Second Card is Clubs	Second card is a clubs	position_second	`first_card`
atomic_039	Second Card is Diamonds	Second card is a diamonds	position_second	`first_card`
atomic_040	Second Card is Hearts	Second card is a hearts	position_second	`first_card`
atomic_041	Second Card is Spades	Second card is a spades	position_second	`first_card`
atomic_042	Second Card is Red	Second card is red	position_second	`first_card`
atomic_043	Second Card is Black	Second card is black	position_second	`first_card`

Range Rules

ID	Name	Description	Category	Acceptable Features
atomic_047	All Low Cards	All cards have rank 7 or lower	range	`max_rank`, `min_rank`, `rank_spread`
atomic_048	All High Cards	All cards have rank 8 or higher	range	`max_rank`, `min_rank`, `rank_spread`
atomic_049	Has Low Card	Contains card with rank 5 or lower	range	`max_rank`, `min_rank`, `rank_spread`
atomic_050	Has High Card	Contains card with rank 10 or higher	range	`max_rank`, `min_rank`, `rank_spread`

Level 2: Comparison Rules (22 rules)

Rules comparing properties between positions or halves.

Terminal Comparison Rules

ID	Name	Description	Category	Acceptable Features
compare_051	First and Last Same Suit	First and last cards have the same suit	terminals	`first_card`, `last_card`, `ends_same_suit`, `ends_same_color`
compare_052	First and Last Same Color	First and last cards have the same color	terminals	`first_card`, `last_card`, `ends_same_suit`, `ends_same_color`
compare_053	First and Last Same Rank	First and last cards have the same rank	terminals	`first_card`, `last_card`, `ends_same_suit`, `ends_same_color`
compare_054	First and Last Same Parity	First and last cards have the same rank parity	terminals	`first_card`, `last_card`, `ends_same_suit`, `ends_same_color`
compare_055	First and Last Different Suits	First and last cards have different suits	terminals	`first_card`, `last_card`, `ends_same_suit`, `ends_same_color`
compare_056	First and Last Different Colors	First and last cards have different colors	terminals	`first_card`, `last_card`, `ends_same_suit`, `ends_same_color`
compare_057	First Rank Greater Than Last	First card has higher rank than last card	terminals_rank	`first_card`, `last_card`, `rank_spread`
compare_058	First Rank Less Than Last	First card has lower rank than last card	terminals_rank	`first_card`, `last_card`, `rank_spread`

Halves Comparison Rules

ID	Name	Description	Category	Acceptable Features
compare_059	Halves Uniform Color Equal	Both halves are uniformly colored, or both are not	halves_property	`is_uniform_color`, `halves_same_color`
compare_060	Halves Uniform Suit Equal	Both halves have uniform suit, or both don't	halves_property	`is_uniform_color`, `halves_same_color`
compare_061	Halves Same Suit Set	Both halves contain the same set of suits	halves_set	`unique_suits`, `halves_same_color`
compare_062	Halves Same Color Set	Both halves contain the same set of colors	halves_set	`unique_suits`, `halves_same_color`
compare_063	Left Half Sum Greater	Sum of ranks in left half is greater than right half	halves_sum	`sum_ranks`
compare_064	Halves Equal Sum	Sum of ranks in left half equals right half	halves_sum	`sum_ranks`

Adjacent Comparison Rules

ID	Name	Description	Category	Acceptable Features
compare_065	All Adjacent Same Color	All adjacent cards have the same color	adjacent	`is_sorted`, `rank_spread`
compare_066	All Adjacent Different Colors	All adjacent cards have different colors (alternating)	adjacent	`is_sorted`, `rank_spread`

Position Pair Rules

ID	Name	Description	Category	Acceptable Features
compare_067	Position 0 and 1 Same Suit	Cards at positions 0 and 1 have the same suit	position_pair	`first_card`, `last_card`, `ends_same_suit`, `ends_same_color`
compare_068	Position 0 and 2 Same Suit	Cards at positions 0 and 2 have the same suit	position_pair	`first_card`, `last_card`, `ends_same_suit`, `ends_same_color`
compare_069	Position 1 and 2 Same Suit	Cards at positions 1 and 2 have the same suit	position_pair	`first_card`, `last_card`, `ends_same_suit`, `ends_same_color`
compare_070	Position 1 and 3 Same Suit	Cards at positions 1 and 3 have the same suit	position_pair	`first_card`, `last_card`, `ends_same_suit`, `ends_same_color`

ID	Name	Description	Category	Acceptable Features
compare_071	Position 2 and 3 Same Suit	Cards at positions 2 and 3 have the same suit	position_pair	`first_card`, `last_card`, `ends_same_suit`, `ends_same_color`
compare_072	Position 2 and 4 Same Suit	Cards at positions 2 and 4 have the same suit	position_pair	`first_card`, `last_card`, `ends_same_suit`, `ends_same_color`

Level 3: Counting Rules (47 rules)

Rules based on counting elements.

Unique Count Rules

ID	Name	Description	Category	Acceptable Features
count_073	Exactly 1 Unique Suit	Hand contains exactly 1 different suit	unique_count	`unique_suits`, `unique_colors`, `unique_ranks`
count_074	Exactly 2 Unique Suits	Hand contains exactly 2 different suits	unique_count	`unique_suits`, `unique_colors`, `unique_ranks`
count_075	Exactly 3 Unique Suits	Hand contains exactly 3 different suits	unique_count	`unique_suits`, `unique_colors`, `unique_ranks`
count_076	Exactly 4 Unique Suits	Hand contains exactly 4 different suits	unique_count	`unique_suits`, `unique_colors`, `unique_ranks`
count_077	Exactly 1 Unique Color	Hand contains exactly 1 different color	unique_count	`unique_suits`, `unique_colors`, `unique_ranks`
count_078	Exactly 2 Unique Colors	Hand contains exactly 2 different colors	unique_count	`unique_suits`, `unique_colors`, `unique_ranks`
count_079	Exactly 2 Unique Ranks	Hand contains exactly 2 different ranks	unique_count	`unique_suits`, `unique_colors`, `unique_ranks`
count_080	Exactly 3 Unique Ranks	Hand contains exactly 3 different ranks	unique_count	`unique_suits`, `unique_colors`, `unique_ranks`
count_081	Exactly 4 Unique Ranks	Hand contains exactly 4 different ranks	unique_count	`unique_suits`, `unique_colors`, `unique_ranks`
count_082	Exactly 5 Unique Ranks	Hand contains exactly 5 different ranks	unique_count	`unique_suits`, `unique_colors`, `unique_ranks`
count_083	Exactly 6 Unique Ranks	Hand contains exactly 6 different ranks	unique_count	`unique_suits`, `unique_colors`, `unique_ranks`

Unique Bound Rules

ID	Name	Description	Category	Acceptable Features
count_084	At Most 2 Unique Suits	Hand contains at most 2 different suits	unique_bound	`unique_suits`, `unique_colors`, `unique_ranks`
count_085	At Most 3 Unique Suits	Hand contains at most 3 different suits	unique_bound	`unique_suits`, `unique_colors`, `unique_ranks`
count_086	At Least 2 Unique Suits	Hand contains at least 2 different suits	unique_bound	`unique_suits`, `unique_colors`, `unique_ranks`
count_087	At Least 3 Unique Suits	Hand contains at least 3 different suits	unique_bound	`unique_suits`, `unique_colors`, `unique_ranks`
count_088	At Least 4 Unique Suits	Hand contains at least 4 different suits	unique_bound	`unique_suits`, `unique_colors`, `unique_ranks`

Specific Count Rules

ID	Name	Description	Category	Acceptable Features
count_089	Exactly 1 Clubs	Hand contains exactly 1 clubs	count_specific	`suit_count`, `color_count`, `rank_count`
count_090	Exactly 2 Clubs	Hand contains exactly 2 clubs	count_specific	`suit_count`, `color_count`, `rank_count`
count_091	Exactly 3 Clubs	Hand contains exactly 3 clubs	count_specific	`suit_count`, `color_count`, `rank_count`
count_092	Exactly 1 Diamonds	Hand contains exactly 1 diamonds	count_specific	`suit_count`, `color_count`, `rank_count`
count_093	Exactly 2 Diamonds	Hand contains exactly 2 diamonds	count_specific	`suit_count`, `color_count`, `rank_count`
count_094	Exactly 3 Diamonds	Hand contains exactly 3 diamonds	count_specific	`suit_count`, `color_count`, `rank_count`
count_095	Exactly 1 Hearts	Hand contains exactly 1 hearts	count_specific	`suit_count`, `color_count`, `rank_count`
count_096	Exactly 2 Hearts	Hand contains exactly 2 hearts	count_specific	`suit_count`, `color_count`, `rank_count`
count_097	Exactly 3 Hearts	Hand contains exactly 3 hearts	count_specific	`suit_count`, `color_count`, `rank_count`
count_098	Exactly 1 Spades	Hand contains exactly 1 spades	count_specific	`suit_count`, `color_count`, `rank_count`
count_099	Exactly 2 Spades	Hand contains exactly 2 spades	count_specific	`suit_count`, `color_count`, `rank_count`
count_100	Exactly 3 Spades	Hand contains exactly 3 spades	count_specific	`suit_count`, `color_count`, `rank_count`
count_101	Exactly 1 Red Card	Hand contains exactly 1 red card	count_specific	`suit_count`, `color_count`, `rank_count`
count_102	Exactly 2 Red Cards	Hand contains exactly 2 red cards	count_specific	`suit_count`, `color_count`, `rank_count`
count_103	Exactly 3 Red Cards	Hand contains exactly 3 red cards	count_specific	`suit_count`, `color_count`, `rank_count`
count_104	Exactly 4 Red Cards	Hand contains exactly 4 red cards	count_specific	`suit_count`, `color_count`, `rank_count`

ID	Name	Description	Category	Acceptable Features
count_105	Exactly 1 Black Card	Hand contains exactly 1 black card	count_specific	`suit_count`, `color_count`, `rank_count`
count_106	Exactly 2 Black Cards	Hand contains exactly 2 black cards	count_specific	`suit_count`, `color_count`, `rank_count`
count_107	Exactly 3 Black Cards	Hand contains exactly 3 black cards	count_specific	`suit_count`, `color_count`, `rank_count`
count_108	Exactly 4 Black Cards	Hand contains exactly 4 black cards	count_specific	`suit_count`, `color_count`, `rank_count`

Count Comparison Rules

ID	Name	Description	Category	Acceptable Features
count_109	More Red Than Black	More red cards than black cards	count_compare	`suit_count`, `color_count`
count_110	More Black Than Red	More black cards than red cards	count_compare	`suit_count`, `color_count`
count_111	Equal Red and Black	Equal number of red and black cards	count_compare	`suit_count`, `color_count`

Majority Rules

ID	Name	Description	Category	Acceptable Features
count_112	Majority Same Suit	More than half the cards share the same suit	majority	`suit_count`, `color_count`, `is_uniform_color`
count_113	Majority Same Color	More than half the cards share the same color	majority	`suit_count`, `color_count`, `is_uniform_color`

Parity Count Rules

ID	Name	Description	Category	Acceptable Features
count_114	Exactly 1 Odd Rank	Exactly 1 card with odd rank	count_parity	`suit_count`, `color_count`
count_115	Exactly 2 Odd Ranks	Exactly 2 cards with odd rank	count_parity	`suit_count`, `color_count`
count_116	Exactly 3 Odd Ranks	Exactly 3 cards with odd rank	count_parity	`suit_count`, `color_count`

ID	Name	Description	Category	Acceptable Features
count_117	Exactly 1 Even Rank	Exactly 1 card with even rank	count_parity	`suit_count`, `color_count`
count_118	Exactly 2 Even Ranks	Exactly 2 cards with even rank	count_parity	`suit_count`, `color_count`
count_119	Exactly 3 Even Ranks	Exactly 3 cards with even rank	count_parity	`suit_count`, `color_count`

Level 4: Pattern Rules (29 rules)

Rules detecting sequential or structural patterns.

Sorted Rules

ID	Name	Description	Category	Acceptable Features
pattern_120	Sorted by Rank (Non-decreasing)	Ranks are in non-decreasing order left to right	sorted	`is_sorted`
pattern_121	Sorted by Rank (Strictly Increasing)	Ranks are in strictly increasing order	sorted	`is_sorted`
pattern_122	Sorted by Rank (Non-increasing)	Ranks are in non-increasing order	sorted	`is_sorted`
pattern_148	Monotonic Ranks	Ranks are either non-decreasing or non-increasing	monotonic	`is_sorted`, `rank_spread`

Alternating Rules

ID	Name	Description	Category	Acceptable Features
pattern_123	Alternating Colors	Colors alternate (red-black-red-black…)	alternating	`is_uniform_color`, `unique_colors`
pattern_124	Alternating Parities	Rank parities alternate (odd-even-odd-even…)	alternating	`is_uniform_color`, `unique_colors`

Duplicates Rules

ID	Name	Description	Category	Acceptable Features
pattern_125	Has Pair (Same Rank)	At least two cards share the same rank	duplicates	`has_pair`, `has_triple`, `unique_ranks`
pattern_126	Has Triple (Three of a Kind)	At least three cards share the same rank	duplicates	`has_pair`, `has_triple`, `unique_ranks`
pattern_127	Has Adjacent Suit Pair	At least two adjacent cards share the same suit	duplicates	`has_pair`, `has_triple`, `unique_ranks`

Palindrome Rules

ID	Name	Description	Category	Acceptable Features
pattern_128	Suits Palin-drome	Suit sequence reads the same forwards and backwards	palindrome	`is_palindrome_colors`, `is_palindrome_suits`
pattern_129	Colors Palin-drome	Color sequence reads the same forwards and backwards	palindrome	`is_palindrome_colors`, `is_palindrome_suits`
pattern_130	Ranks Palin-drome	Rank sequence reads the same forwards and backwards	palindrome	`is_palindrome_colors`, `is_palindrome_suits`
pattern_131	Parities Palin-drome	Parity sequence reads the same forwards and backwards	palindrome	`is_palindrome_colors`, `is_palindrome_suits`

Halves Copy Rules

ID	Name	Description	Category	Acceptable Features
pattern_132	Halves Copy Suits	Left and right halves have the same suit sequence	halves_copy	`is_palindrome_suits`, `halves_same_color`
pattern_133	Halves Copy Colors	Left and right halves have the same color sequence	halves_copy	`is_palindrome_suits`, `halves_same_color`
pattern_134	Halves Copy Ranks	Left and right halves have the same rank sequence	halves_copy	`is_palindrome_suits`, `halves_same_color`

Run Rules

ID	Name	Description	Category	Acceptable Features
pattern_135	Has Run of 3	Contains at least 3 consecutive ranks	runs	`is_sorted`, `rank_spread`
pattern_136	Has Run of 4	Contains at least 4 consecutive ranks	runs	`is_sorted`, `rank_spread`

Adjacent Constraint Rules

ID	Name	Description	Category	Acceptable Features
pattern_137	Adjacent Rank Gap <= 2	All adjacent cards differ by at most 2 in rank	adjacent_constraint	`is_sorted`, `unique_suits`, `unique_colors`
pattern_138	Adjacent Rank Gap <= 3	All adjacent cards differ by at most 3 in rank	adjacent_constraint	`is_sorted`, `unique_suits`, `unique_colors`
pattern_139	Adjacent Same Rank or Suit	All adjacent cards share either rank or suit	adjacent_constraint	`is_sorted`, `unique_suits`, `unique_colors`

Shape Rules

ID	Name	Description	Category	Acceptable Features
pattern_140	V-Shape Ranks	Ranks descend to middle then ascend (V shape)	shape	`has_pair`, `has_triple`, `unique_ranks`
pattern_141	Inverted V-Shape Ranks	Ranks ascend to middle then descend (inverted V)	shape	`has_pair`, `has_triple`, `unique_ranks`

Periodic Rules

ID	Name	Description	Category	Acceptable Features
pattern_142	Period-2 Colors	Colors repeat with period 2 (ABABAB...)	periodic	`unique_suits`, `unique_colors`
pattern_143	Period-2 Suits	Suits repeat with period 2 (ABABAB...)	periodic	`unique_suits`, `unique_colors`

Skip Rules

ID	Name	Description	Category	Acceptable Features
pattern_144	Skip-1 Same Suit	Cards at positions i and i+2 have same suit	skip	`is_sorted`, `rank_spread`
pattern_145	Skip-1 Same Color	Cards at positions i and i+2 have same color	skip	`is_sorted`, `rank_spread`

Spread Rules

ID	Name	Description	Category	Acceptable Features
pattern_146	Rank Spread >= 8	Difference between highest and lowest rank is at least 8	spread	`max_rank`, `min_rank`, `rank_spread`
pattern_147	Rank Spread <= 5	Difference between highest and lowest rank is at most 5	spread	`max_rank`, `min_rank`, `rank_spread`

Level 5: Compositional Rules (53 rules)

Rules combining multiple conditions with logical connectives.

AND Combination Rules

ID	Name	Description
comp_149	All Same Color AND Sorted	All cards same color AND ranks are sorted
comp_150	Has Pair AND Alternating Colors	Has a pair AND colors alternate
comp_151	First Red AND Last Black	First card is red AND last card is black
comp_152	First Black AND Last Red	First card is black AND last card is red
comp_153	Exactly 2 Suits AND Has Pair	Exactly 2 suits AND has at least one pair
comp_154	Sum Even AND First Red	Sum of ranks is even AND first card is red
comp_155	Halves Same Colors AND Sorted	Both halves have same color set AND ranks are sorted
comp_171	Alternating Colors AND First Spade	Colors alternate AND first card is a spade
comp_172	Has Run of 3 AND Uniform Color	Contains 3 consecutive ranks AND all cards same color
comp_173	Palindrome Colors AND Has Heart	Color sequence is palindrome AND has at least one heart
comp_174	More Black AND Last Black	More black cards than red AND last card is black
comp_175	At Least 3 Suits AND Sorted	At least 3 different suits AND ranks are sorted
comp_192	Has Heart AND Has Club	Has at least one heart AND at least one club
comp_193	Has Diamond AND Has Spade	Has at least one diamond AND at least one spade
comp_194	All Four Suits Present	Hand contains cards from all four suits
comp_195	Exactly One Suit Missing	Exactly three suits are present (one is missing)
comp_197	First Pair AND Last Pair Same Suit	First two cards share suit AND last two cards share suit
comp_200	Sum Div 3 AND Has Heart	Sum of ranks divisible by 3 AND has at least one heart
comp_201	Halves Copy Colors AND First Black	Left and right halves have same color sequence AND first card is black

OR Combination Rules

ID	Name	Description
comp_156	All Same Suit OR All Same Color	All cards same suit OR all cards same color
comp_157	Has Pair OR Sorted	Has a pair OR ranks are sorted
comp_158	First Red OR Last Red	First card is red OR last card is red
comp_159	Palindrome Suits OR Palindrome Colors	Suits form palindrome OR colors form palindrome
comp_160	Sum Even OR Has Spade	Sum of ranks is even OR has at least one spade
comp_176	Has Triple OR All Same Color	Has three of a kind OR all cards same color
comp_177	First Black OR Last Black	First card is black OR last card is black
comp_178	Sum Even OR Alternating Colors	Sum of ranks is even OR colors alternate
comp_179	More Suit Variety OR Has Pair	More unique suits than colors OR has a pair
comp_196	All Even OR All Odd Ranks	All cards have even ranks OR all cards have odd ranks
comp_198	First Pair OR Last Pair Same Color	First two cards share color OR last two cards share color

Conditional Rules

ID	Name	Description
comp_161	If First Red Then Last Black	If first card is red, then last card must be black
comp_162	If Has Spade Then Has Heart	If hand has a spade, it must also have a heart
comp_163	If Sorted Then Has Pair	If ranks are sorted, then must have at least one pair
comp_164	If Uniform Color Then Not Sorted	If all cards same color, then ranks must not be sorted
comp_180	If Uniform Color Then Sum Even	If all cards same color, then sum of ranks must be even
comp_181	If Has Pair Then Not Palindrome Suits	If hand has a pair, then suits must not form palindrome
comp_182	If First Heart Then Last Diamond	If first card is heart, then last card must be diamond
comp_183	If More Red Then First Red	If more red than black, then first card must be red

XOR Rules

ID	Name	Description
comp_169	Same Suit XOR Same Color	All same suit XOR all same color (exactly one, not both)
comp_170	Sorted XOR Has Pair	Sorted XOR has pair (exactly one, not both)

Negation Rules

ID	Name	Description
comp_186	Not All Same Suit	Not all cards have the same suit (at least 2 different suits)
comp_187	Not Sorted	Ranks are NOT in non-decreasing order
comp_188	Not Palindrome Colors	Color sequence is NOT a palindrome
comp_191	Neither Same Suit Nor Same Color	Not all same suit AND not all same color
comp_199	No Pair AND Not Sorted	No pairs AND ranks are not sorted

Complex Rules

ID	Name	Description
comp_165	Opposite Terminal Colors	First and last cards have opposite colors
comp_166	Same Color AND (Sorted OR Pair)	All same color AND (sorted OR has pair)
comp_167	Has Both of a Color Pair	Has (spade AND heart) OR (diamond AND club)
comp_168	More Red AND First Red	More red cards than black AND first card is red

Triple Combination Rules

ID	Name	Description
comp_184	Same Color AND Sorted AND Pair	All same color AND sorted AND has at least one pair
comp_185	First Red AND Last Black AND Has Spade	First is red AND last is black AND has at least one spade

Biconditional Rules

ID	Name	Description
comp_189	Same Suit IFF Same Color	All same suit if and only if all same color
comp_190	Has Pair IFF <= 5 Unique Ranks	Has pair if and only if at most 5 unique ranks

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Part 5: Evaluation Results Summary

From the quick evaluation (December 26, 2024):

Metric	Value
Semantic Correctness	**52.9%**
Consistency	94.1%
Discrimination	98.9%
FCR@3 (legacy)	0%

By Level

Level	Description	Semantic Correctness
1	Atomic	40.0%
2	Comparison	66.7%
3	Counting	83.3%
4	Pattern	40.0%
5	Compositional	50.0% (lenient)

Part 6: How Semantic Correctness Works

The Evaluation Algorithm

```
For each rule:
  1. Get the rule's category (e.g., "uniform", "sorted")
  2. Look up acceptable features for that category
  3. Generate descriptions for task examples
  4. For each description in top-3:
     a. Extract features from description object
     b. Extract features from description text (keyword matching)
     c. Check if any feature overlaps with acceptable set
  5. Score = 1.0 if ANY description matches, else 0.0
```

Example

Rule: "All Same Suit" (category: `uniform`) **Acceptable features**: `is_flush`, `is_uniform_color`, `unique_suits`, `unique_colors`, `unique_ranks`

Generated description: "all cards share the same suit" **Extracted features**: `is_flush` (from direct feature), `unique_suits` (from "same suit" text) **Overlap with acceptable**: `is_flush` matches! **Score**: 1.0 (semantically correct)

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